

MANJUSHA K

Bangalore, Karnataka • +91 9449949910 • manjushamukundan31@gmail.com
linkedin.com/in/manjusha-k-mukundan-64642a26a • GitHub: Mxnjusha

PROFESSIONAL SUMMARY

Data Engineer with a strong foundation in Statistics, Big Data Analytics, and cloud-based data infrastructure. Experienced in designing ETL/ELT pipelines, building analytics-ready datasets, and delivering business intelligence solutions using Python, PySpark, SQL, Power BI, and AWS. Demonstrated ability to transform raw enterprise data into operational insights supporting planning, fraud detection, and clinical decision-making. Eager to modernize legacy data workflows and drive retail analytics outcomes in a cloud-native Databricks environment.

TECHNICAL SKILLS

Data Engineering & Pipelines: ETL/ELT Workflows, Data Pipelines, Data Cleaning & Wrangling, Document Parsing, OCR, Data Validation & Reconciliation

Languages & Frameworks: Python (Pandas, NumPy, scikit-learn, TensorFlow, Keras), PySpark, R, SQL, Flask, FastAPI

Cloud & DevOps: AWS (EC2, S3, Terraform, AWS CLI), Docker, Git, CI/CD practices, Streamlit, HuggingFace Spaces

BI & Visualization: Power BI, Tableau, Matplotlib, KPI Tracking, Dashboard Development, Data Storytelling

Machine Learning & Analytics: Supervised/Unsupervised Learning, Random Forest, Logistic Regression, SVM, CNNs, A/B Testing, Time Series Forecasting, Regression Analysis

Data Modeling: Star & Snowflake Schema concepts, Retail KPI datasets (sales, inventory, demand), Analytics-ready dataset design

Additional Technologies: LangChain, LlamaIndex, Pinecone, OpenAI APIs, Supabase, n8n, Flowise, RAG Pipelines

EXPERIENCE

AI / GenAI Engineer Intern | Trigent Software Pvt Ltd

Jan 2026 – Mar 2026
Bengaluru, India

- Designed, built, and validated ETL workflows for structured and unstructured enterprise datasets, ensuring data quality, reliability, and operational readiness for analytics and reporting applications.
- Built interactive Power BI-style dashboards and reporting solutions for fraud detection and operational monitoring, enabling data-driven decision-making across cross-functional stakeholders.
- Developed and deployed ML-based fraud detection system (Random Forest, Logistic Regression) achieving 95%+ prediction accuracy with real-time Streamlit dashboards, directly reducing false positives in transaction monitoring.
- Engineered healthcare data pipelines for medication safety, renal dose adjustment, and clinical risk assessment — integrating multiple data sources (Supabase, n8n) into live clinical environments.
- Designed and deployed production-grade data pipelines using document chunking, embedding generation, and vector databases (Pinecone), enabling sub-second knowledge retrieval across enterprise document stores.
- Validated and debugged 800+ enterprise automation workflows, improving data quality, system reliability, and cross-platform integration performance across production pipelines.
- Developed clinical monitoring platforms including chemotherapy toxicity surveillance and dialysis risk scoring; generated actionable alerts through analytics dashboards for healthcare staff.
- Built an intelligent Rare Disease Diagnostic system using OCR-based clinical data extraction and HPO symptom mapping — an end-to-end data pipeline from unstructured inputs to classification outputs.

PROJECTS

EV Adoption & Air Quality Analysis — Electrifying Bengaluru

Tools: Python, Pandas, NumPy, Matplotlib, scikit-learn, Excel

- Analyzed six years (2018–2024) of large-scale EV adoption and air quality data — performing data cleaning, preprocessing, and exploratory analysis to identify trends and quantify environmental impact.

- Designed forecasting models projecting EV penetration and PM2.5 reduction under different policy scenarios; translated findings into infrastructure planning recommendations via data visualizations and reports.

A/B Testing — E-Commerce Landing Page Conversion Optimization

Tools: Python, Pandas, Statsmodels, Matplotlib, Excel

- Designed and executed statistically rigorous A/B tests (two-proportion z-tests) to evaluate landing page performance; cleaned and analyzed user behavior data to measure conversion improvements.
- Delivered actionable insights and visualizations that guided product team's data-driven redesign decisions — a direct parallel to retail KPI analysis and planning cycle support.

Handwriting-Based Parkinson's Disease Detection

Tools: Python, TensorFlow, Keras, scikit-learn, ResNet50, SVM, Flask

- Engineered a ResNet50 + SVM hybrid deep learning pipeline achieving 97.47% accuracy (AUC 0.989) via stratified 5-fold cross-validation; applied transfer learning and data augmentation to expand dataset from 3,500 to 7,100+ images.
- Shipped model as a Flask web application enabling real-time diagnostic inference — demonstrating end-to-end data product delivery from raw inputs to production deployment.

Cloud-Deployed Appointment Booking Web App

Tools: Python, Flask, SQLAlchemy, SQLite, AWS EC2, Git

- Developed a full-stack application with database management (SQLAlchemy ORM), payment gateway integration, and automated email notifications; deployed on AWS EC2 with production-ready configuration.

EDUCATION

M.Sc. in Big Data Analytics | St. Joseph's University, Bengaluru

Jul 2024 – May 2025

SGPA: 8.06 • Coursework: Big Data, Machine Learning, Statistical Modelling, Data Engineering

B.Sc. in Statistics & Mathematics | Mount Carmel College, Bengaluru

Aug 2021 – Apr 2024

SGPA: 6.91

CERTIFICATIONS & LEADERSHIP

- Google Analytics Certification — Great Learning
- Goldman Sachs Excel Skills for Business — Forage
- Head of Media & Marketing Committee — Datagram 7.0, St. Joseph's University
- Core Team Member — Reel Run Association & Mathematics Association, Mount Carmel College